

Que. 1: What are the different protocols you observe at the following layers of the protocol stack?

- A. Application layer
- B. Transport layer
- C. Network layer

Ans:- During the packet capture in Wireshark while accessing `http://neverssl.com` and `https://www.cornell.edu`, the following protocols were observed at different layers of the protocol stack.

Application layer -

The following application layer protocols were observed:-

- HTTP (Hypertext Transfer Protocol) -

- It is used to access web pages and transfer data b/w the client and the server.

- DNS (Domain Name System)

- used to resolve the domain names into IP addresses before establishing the connection.

- TLS (Transport Layer Security)

- used to establish a secure encrypted connection while accessing `https://www.cornell.edu`.

Transport Layer

The following transport layer protocols were observed.

- TCP (Transmission control Protocol)! -
used to provide reliable communication for HTTP and HTTPS traffic.
- UDP (user Datagram protocol)! -
 - used mainly by DNS queries and responses because it is faster and suitable for small data transfers.

Network Layer -

The following network layer protocol was observed.

- IPv4 (Internet protocol version - 4)! -
 - used to route packets b/w source and destination IP addresses over the network.

No.	Time	Source	Destination	Protocol	Length	Info
5620	39.238277000	10.85.20.254	104.192.109.75	HTTP	588	POST /commonquery HTTP/1.1 (application/x-www-form-urlencoded)
5621	39.240175200	10.85.20.254	104.192.109.75	HTTP	580	POST /commonquery HTTP/1.1 (application/x-www-form-urlencoded)
5664	39.511875400	104.192.109.75	10.85.20.254	TCP	1514	[TCP Previous segment not captured] 80 → 49175 [PSH, ACK] Seq=1461 Ack=535 Win=64128 Len=14...
5665	39.511878900	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5667	39.511887400	104.192.109.75	10.85.20.254	HTTP	1514	[TCP Previous segment not captured] Continuation
5668	39.511889200	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5669	39.511891300	104.192.109.75	10.85.20.254	HTTP	1514	[TCP Out-Of-Order] Continuation
5670	39.511893200	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5671	39.511895600	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5672	39.511900900	104.192.109.75	10.85.20.254	HTTP	2974	Continuation
5686	39.533445400	104.192.109.75	10.85.20.254	HTTP	4434	[TCP Previous segment not captured] Continuation
5689	39.533447400	104.192.109.75	10.85.20.254	HTTP	1514	[TCP Out-Of-Order] Continuation
5690	39.533450700	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5709	39.787608100	104.192.109.75	10.85.20.254	HTTP	1514	Continuation

> Frame 5620: Packet, 588 bytes on wire (4704 bits), 588 bytes captured (4704 bits) on
 > Ethernet II, Src: Intel_f2:f8:b9 (Sc:80:b6:f2:f8:b9), Dst: ASIXElectron_35:2d:a0 (fi
 > Internet Protocol Version 4, Src: 10.85.20.254, Dst: 104.192.109.75
 > Transmission Control Protocol, Src Port: 49175, Dst Port: 80, Seq: 1, Ack: 1, Len: !

▼ Hypertext Transfer Protocol

> POST /commonquery HTTP/1.1\r\n
 Host: spec.cloud.360safe.com\r\n
 Accept: */*\r\n
 Connection: Keep-Alive\r\n
 Cache-Control: no-cache\r\n
 > Content-Length: 342\r\n
 Content-Type: application/x-www-form-urlencoded\r\n
 \r\n
 [Full request URI: http://spec.cloud.360safe.com/commonquery]

```

0070 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 43 6f 6e Accept: */* Con
0080 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 6c nection: Keep-Al
0090 69 76 65 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72 ive..Cac he-Contr
00a0 6f 6c 3a 20 6e 6f 2d 63 61 63 68 65 0d 0a 43 6f ol: no-c ache..Co
00b0 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 33 34 ntent-Le ngth: 34
00c0 32 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 2..Conte nt-Type:
00d0 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 2d 77 applica tion/x-w
00e0 77 77 2d 66 6f 72 6d 2d 75 72 6c 65 6e 63 6f 64 ww-form- urlencod
00f0 65 64 0d 0a 0d 0a 0e 0e 24 21 04 08 ef 64 d5 18 ed..... $!...d...
0100 00 00 00 00 f7 91 ec bd 45 aa cd ed 5f 54 51 3c ..... E...TQC
0110 55 63 0d 19 02 b3 de 14 fe 3b e7 3c af 54 45 77 Uc..... ;< TEW
0120 5d f8 b9 88 84 a0 17 d9 92 c5 c4 76 f7 dc c5 3d ]..... v...=
0130 03 47 07 a6 0e e6 c4 a2 8e 50 c8 a8 e4 fd 1b 6d .G..... P...m
0140 7a 8e 6e 56 b0 b6 3b 0e a6 84 90 f7 34 e9 b7 f6 z-nV...;...4...
0150 e4 23 0e ef 48 63 8d 26 2d ac 9b 29 ad bc 8f da .#..Mc & ...).
0160 ca bc ec 55 d5 8b a3 68 55 f3 ff e2 ed 28 e6 29 ..U..h U...().
0170 84 33 d1 68 34 c1 2a 89 bc 0c bb 86 9f 2d 73 2c .3.h4... ..s,
0180 16 f2 ca df 65 9b ee 04 05 06 18 0c 17 42 b4 bb ...e..... B
    
```

HTTP Accept (http.accept). 13 bytes

Packets: 16301 - Displayed: 62 (0.4%) - Dropped: 0 (0.0%) Profile: Default



tcp

No.	Time	Source	Destination	Protocol	Length	Info
5595	38.928600000	57.155.142.104	10.85.20.254	TCP	66	443 → 49174 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5596	38.928927500	10.85.20.254	57.155.142.104	TCP	54	49174 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
5616	39.219027000	104.192.109.75	10.85.20.254	TCP	62	80 → 49175 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=128
5617	39.219028300	104.192.109.75	10.85.20.254	TCP	62	80 → 49176 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=128
5618	39.220895600	10.85.20.254	104.192.109.75	TCP	54	49175 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
5619	39.221141500	10.85.20.254	104.192.109.75	TCP	54	49176 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
5620	39.238277000	10.85.20.254	104.192.109.75	HTTP	588	POST /commonquery HTTP/1.1 (application/x-www-form-urlencoded)
5621	39.240175200	10.85.20.254	104.192.109.75	HTTP	580	POST /commonquery HTTP/1.1 (application/x-www-form-urlencoded)
5622	39.327026800	10.85.20.254	57.155.142.104	TLSv1.2	522	Client Hello (SNI=wdcp.microsoft.com)
5630	39.357224400	10.85.20.254	162.159.61.3	TCP	54	54263 → 443 [FIN, ACK] Seq=2 Ack=1 Win=253 Len=0
5631	39.357968300	10.85.20.254	192.178.158.84	TCP	54	56991 → 443 [FIN, ACK] Seq=1810 Ack=6381 Win=64512 Len=0
5632	39.358369000	10.85.20.254	142.250.122.102	TCP	54	59593 → 443 [FIN, ACK] Seq=1828 Ack=6771 Win=64512 Len=0
5633	39.358987000	10.85.20.254	104.18.20.122	TCP	54	64336 → 443 [FIN, ACK] Seq=1 Ack=1 Win=255 Len=0
5634	39.359265000	10.85.20.254	1.1.1.1	TCP	54	52454 → 443 [FIN, ACK] Seq=1 Ack=1 Win=252 Len=0

> Frame 5620: Packet, 588 bytes on wire (4704 bits), 588 bytes captured (4704 bits) on
 > Ethernet II, Src: Intel_f2:f8:b9 (5c:80:b6:f2:f8:b9), Dst: ASIXElectron_35:2d:a0 (f1
 > Internet Protocol Version 4, Src: 10.85.20.254, Dst: 104.192.109.75
 > Transmission Control Protocol, Src Port: 49175, Dst Port: 80, Seq: 1, Ack: 1, Len: !
 > Hypertext Transfer Protocol
 > POST /commonquery HTTP/1.1\r\n
 Host: spec.cloud.360safe.com\r\n
 Accept: */*\r\n
 Connection: Keep-Alive\r\n
 Cache-Control: no-cache\r\n
 > Content-Length: 342\r\n
 Content-Type: application/x-www-form-urlencoded\r\n
 \r\n
 [Full request URI: http://spec.cloud.360safe.com/commonquery]

```

0090 69 76 65 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00a0 6f 6c 3a 20 6e 6f 2d 63 61 63 68 65 0d 0a 43 6f  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00b0 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 33 34  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00c0 32 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00d0 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 2d 77  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00e0 77 77 2d 66 6f 72 6d 2d 75 72 6c 65 6e 63 6f 64  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00f0 65 64 0d 0a 0d 0a 0e 06 24 21 04 08 ef 64 d5 18  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0100 00 00 09 00 f7 91 ec bd 45 aa cd ed 5f 54 51 3c  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0110 55 63 0d 19 02 b3 de 14 fe 3b e7 3c af 54 45 77  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0120 5d f8 b9 88 8a a0 17 d9 92 c5 4a 76 f7 dc c5 3d  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0130 03 47 07 a6 0e e6 c4 a2 8e 50 c8 a8 e4 fd 1b 6d  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0140 7a 8e 6e 56 b0 b6 3b 0e a6 84 90 f7 34 e9 b7 f6  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0150 e4 23 0e ef 48 63 8d 26 2d ac 9b 29 ad bc 8f da  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0160 ca bc ec 55 d5 8b a3 68 55 f3 ff e2 ed 28 e6 29  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0170 84 33 d1 68 34 c1 2a 89 bc 0c bb 86 9f 2d 73 2c  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0180 16 f2 ca df 65 9b ee 04 05 06 18 0c 17 42 b4 bb  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
0190 ba 47 ba 0a 26 51 06 0f 9c d2 b3 fd bc 2a 6d 76  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
01a0 ce 50 f2 af 7b 85 b6 6b 28 68 ec 5f d5 f4 fa dd  00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000

```

HTTP Cache Control (http.cache_control), 25 bytes

Packets: 16301 · Displayed: 4216 (25.9%) · Dropped: 0 (0.0%) · Profiler: Default



dns

No.	Time	Source	Destination	Protocol	Length	Info
5051	33.283338400	1.192.193.155	10.85.20.254	DNS	670	Standard query 0x0c0b Unknown (28973) <Unknown extended label>[Malformed Packet]
5532	38.349720100	10.85.20.254	8.8.8.8	DNS	92	Standard query 0x9447 A reflector.defender.microsoft.com
5547	38.414628400	8.8.8.8	10.85.20.254	DNS	289	Standard query response 0x9447 A reflector.defender.microsoft.com CNAME prod-atm-wds-eastus.
5561	38.624135800	10.85.20.254	8.8.8.8	DNS	78	Standard query 0x4b38 A wdcip.microsoft.com
5570	38.705332900	8.8.8.8	10.85.20.254	DNS	209	Standard query response 0x4b38 A wdcip.microsoft.com CNAME wd-prod-cp.trafficmanager.net CNAME
5571	38.797442800	10.85.20.254	8.8.8.8	DNS	82	Standard query 0xcfff A spec.cloud.360safe.com
5580	38.841292000	8.8.8.8	10.85.20.254	DNS	145	Standard query response 0xcfff A spec.cloud.360safe.com CNAME spec.cloud.os-lb.com A 104.19...
5726	39.828581400	10.85.20.254	8.8.8.8	DNS	73	Standard query 0x4f3c A s.360safe.com
5736	39.874098600	8.8.8.8	10.85.20.254	DNS	105	Standard query response 0x4f3c A s.360safe.com A 54.255.136.181 A 54.254.196.234
5812	40.697391900	10.85.20.254	104.192.108.79	DNS	772	Inverse query 0x0a04[Malformed Packet]
5824	40.777099400	10.85.20.254	8.8.8.8	DNS	85	Standard query 0x9e81 A tsst.360totalsecurity.com
5825	40.847297800	8.8.8.8	10.85.20.254	DNS	101	Standard query response 0x9e81 A tsst.360totalsecurity.com A 8.219.181.32
5852	40.973003900	104.192.108.79	10.85.20.254	DNS	118	Standard query 0x0a04[Malformed Packet]
5854	40.975933200	10.85.20.254	104.192.108.79	DNS	772	Inverse query 0x0a04[Malformed Packet]

- > Frame 5580: Packet, 145 bytes on wire (1160 bits), 145 bytes captured (1160 bits) on 1
- > Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: Intel_f2:f8:b9 (5c:8
- > Internet Protocol Version 4, Src: 8.8.8.8, Dst: 10.85.20.254
- > User Datagram Protocol, Src Port: 53, Dst Port: 57088
- > Domain Name System (response)

```
0000 5c 80 b6 f2 f8 b9 f8 e4 3b 35 2d a0 08 00 45 00  \.....;5...E
0010 00 83 a8 ac 00 00 79 11 69 5b 08 08 08 08 0a 55  \.....y i[....U
0020 14 fe 00 35 df 00 00 6f 79 58 cf ff 81 80 00 01  \...5...o yX....
0030 00 03 00 00 00 00 04 73 70 65 63 05 63 6c 6f 75  \.....s pec-clou
0040 64 07 33 36 30 73 61 66 65 03 63 6f 6d 00 00 01  \d-360saf e.com...
0050 00 01 c0 0c 00 05 00 01 00 00 00 98 00 13 04 73  \.....s
0060 70 65 63 05 63 6c 6f 75 64 05 6f 73 2d 6c 62 c0  \pec-clou d-os-lb-
0070 1f c0 34 00 01 00 01 00 00 00 2a 00 04 68 c0 6d  \-4.....h-m
0080 4b c0 34 00 01 00 01 00 00 00 2a 00 04 68 c0 6c  \x-4.....h-l
0090 98
```

Domain Name System (dns), 103 bytes

Packets: 16301 - Displayed: 150 (0.9%) Dropped: 0 (0.0%)

Profile: Default



tls

No.	Time	Source	Destination	Protocol	Length	Info
5508	38.262705000	192.178.158.188	10.85.20.254	TLSv1.3	1470	[TCP Out-Of-Order] , Server Hello, Change Cipher Spec
5517	38.269588000	10.85.20.254	192.178.158.188	TLSv1.3	118	Change Cipher Spec, Application Data
5518	38.278349800	10.85.20.254	192.178.158.188	TLSv1.3	232	Application Data
5546	38.403737700	192.178.158.188	10.85.20.254	TLSv1.3	595	Application Data, Application Data
5548	38.416809200	192.178.158.188	10.85.20.254	TLSv1.3	151	Application Data
5549	38.416811300	192.178.158.188	10.85.20.254	TLSv1.3	92	Application Data
5577	38.819376400	172.217.116.4	10.85.20.254	TLSv1.2	127	Application Data
5622	39.327026800	10.85.20.254	57.155.142.104	TLSv1.2	522	Client Hello (SNI=wedp.microsoft.com)
5657	39.442973500	57.155.142.104	10.85.20.254	TLSv1.2	1514	[TCP Previous segment not captured] , Ignored Unknown Record
5658	39.442978700	57.155.142.104	10.85.20.254	TLSv1.2	972	Ignored Unknown Record
5696	39.537873000	10.85.20.254	57.155.142.104	TLSv1.2	134	Change Cipher Spec, Application Data
5700	39.701508700	10.85.20.254	57.155.142.104	TLSv1.2	511	Application Data
5702	39.702739000	10.85.20.254	57.155.142.104	TLSv1.2	1290	Application Data
5703	39.708838200	57.155.142.104	10.85.20.254	TLSv1.2	157	Application Data

> Frame 5577: Packet, 127 bytes on wire (1016 bits), 127 bytes captured (1016 bits) on
 > Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: Intel_f2:f8:b9 (5c:8d:93:45:00:00)
 > Internet Protocol Version 4, Src: 172.217.116.4, Dst: 10.85.20.254
 > Transmission Control Protocol, Src Port: 443, Dst Port: 51011, Seq: 1, Ack: 1, Len: 75
 > Transport Layer Security

```

0000 5c 80 b6 f2 f8 b9 f8 e4 3b 35 2d a0 08 00 45 00 \.....;5....E-
0010 00 71 04 10 00 00 79 06 fd 46 ac d9 74 04 0a 55 -q....y..F..t..U
0020 14 fe 01 bb c7 43 20 86 ab 56 23 72 37 46 50 18 .....C..V87FP-
0030 04 11 d7 87 00 00 17 03 03 00 44 0c 26 3c c4 49 .....D.<I
0040 92 f1 69 b5 40 4b 9d 91 04 77 80 0a e2 d2 47 84 ..i.k...w....G-
0050 b4 ce d1 c9 97 fa 92 32 47 d6 9b c5 59 2d 51 18 .....2G...Y-Q-
0060 fb 54 a7 e7 8f c1 36 42 29 e7 0f bc 44 7f 31 52 -T....68)...D-1R
0070 dc 42 22 96 4f ff c8 4c d0 62 b3 e3 15 6c ca -B^O-L-b...1.
    
```

Transport Layer Security: Protocol

Packets: 16301 · Displayed: 1466 (9.0%) · Dropped: 0 (0.0%) Profile: Default

cn_lab02.pcapng

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ip

No.	Time	Source	Destination	Protocol	Length	Info
5553	38.469612500	192.178.158.139	10.85.20.254	QUIC	66	Protected Payload (KP0)
5561	38.624135800	10.85.20.254	8.8.8.8	DNS	78	Standard query 0x4b38 A wdcu.microsoft.com
5563	38.705314400	10.85.19.240	224.0.0.251	MDNS	157	Standard query 0x0000 PTR _companion-link_tcp.local, "QM" question PTR Raj's MacBook Air...
5567	38.705318500	10.85.18.124	224.0.0.251	MDNS	96	Standard query 0x0001 PTR _spotify-social-listening_tcp.local, "QU" question
5570	38.705332900	8.8.8.8	10.85.20.254	DNS	209	Standard query response 0x4b38 A wdcu.microsoft.com CNAME wd-prod-cp.trafficmanager.net CNA...
5571	38.797442800	10.85.20.254	8.8.8.8	DNS	82	Standard query 0xcfff A spec.cloud.360safe.com
5577	38.819376400	172.217.116.4	10.85.20.254	TLSv1.2	127	Application Data
5578	38.820032400	10.85.20.254	172.217.116.4	TCP	54	51011 → 443 [FIN, ACK] Seq=1 Ack=74 Win=255 Len=0
5579	38.825449600	10.85.20.254	57.155.142.104	TCP	66	49174 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5580	38.841292000	8.8.8.8	10.85.20.254	DNS	145	Standard query response 0xcfff A spec.cloud.360safe.com CNAME spec.cloud.360safe.com A 104.19...
5581	38.851708400	10.85.20.254	104.192.109.75	TCP	66	49175 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5582	38.851708500	10.85.20.254	104.192.109.75	TCP	66	49176 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
5583	38.865390000	172.217.116.4	10.85.20.254	TCP	60	443 → 51011 [FIN, ACK] Seq=74 Ack=2 Win=1041 Len=0
5584	38.865647400	10.85.20.254	172.217.116.4	TCP	54	51011 → 443 [ACK] Seq=2 Ack=75 Win=255 Len=0

> Frame 5577: Packet, 127 bytes on wire (1016 bits), 127 bytes captured (1016 bits) on 0

> Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: Intel_f2:f8:b9 (5c:8d:1e:f2:f8:b9)

> Internet Protocol Version 4, Src: 172.217.116.4, Dst: 10.85.20.254

> Transmission Control Protocol, Src Port: 443, Dst Port: 51011, Seq: 1, Ack: 1, Len: 75

✓ Transport Layer Security

[Stream index: 41]

> TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol

0000 5c 80 b6 f2 f8 b9 f8 e4 3b 35 2d a0 08 00 45 00 \.....;5---E-

0010 00 71 04 10 00 00 79 06 fd 46 ac d9 74 04 0a 55 .q...y...F...t..U

0020 14 fe 01 bb c7 43 20 86 ab 56 23 72 37 46 50 18C...Vw7FP

0030 04 11 d7 87 00 00 17 03 03 00 44 0c 26 3c c4 49D&c.I

0040 92 f1 69 b5 40 4b 9d 91 04 77 80 0a e2 d2 47 84 ..i@k...w...G

0050 b4 ce d1 c9 97 fa 92 32 47 d6 9b c5 59 2d 51 182G...Y-Q

0060 fb 54 a7 e7 8f c1 36 42 29 e7 0f bc 44 7f 31 52 .T....6B)...D-IR

0070 dc 42 22 96 4f ff c8 4c d0 62 b3 e3 15 6c ca .B..O..L..b...1.

Internet Protocol Version 4: Protocol

Packets: 16301 · Displayed: 8960 (55.0%) · Dropped: 0 (0.0%) Profile: Default

30°C Mostly cloudy

ENG IN

20:00 02-08-2026

Ques. 2 What is the total amount of data being received for the following two cases?

A. When you access <http://neverssl.com/>

Ans/ Total amount of data received while accessing <http://neverssl.com>
= 2877 bytes

Ques 2

B When you access <https://www.cornell.edu>

Ans/ The total amount of data received with <https://www.cornell.edu>.
= .

• Byte $A \rightarrow B = 130$ bytes

• Byte $B \rightarrow A = 0$ bytes



http.host=="neverssl.com"

No.	Time	Source	Destination	Protocol	Length	Info
9310	69.837555000	10.85.20.254	34.223.124.45	HTTP	130	GET / HTTP/1.1

- > Frame 9310: Packet, 130 bytes on wire (1040 bits), 130 bytes captured (1040 bits) on 3
- > Ethernet II, Src: Intel_f2:f8:b9 (5c:80:b6:f2:f8:b9), Dst: ASIXElectron_35:2d:a0 (f8:e
- > Internet Protocol Version 4, Src: 10.85.20.254, Dst: 34.223.124.45
- > Transmission Control Protocol, Src Port: 61487, Dst Port: 80, Seq: 1, Ack: 1, Len: 76
- ✓ Hypertext Transfer Protocol

GET / HTTP/1.1\r\n

Request Method: GET

Request URI: /

Request Version: HTTP/1.1

Host: neverssl.com\r\n

User-Agent: curl/8.21.0\r\n

Accept: */*\r\n

\r\n

[Full request URI: http://neverssl.com/]

```
0000 f8 e4 3b 35 2d a0 5c 80 b6 f2 f8 b9 08 00 45 00 ...;5--\ .....E-
0010 00 74 3f 30 40 00 80 06 00 00 0a 55 14 fe 22 df ..t?@... ..U..-
0020 7c 2d f0 2f 00 50 a3 9e 41 4f 85 d5 e4 7f 50 18 |../P.. AO....P-
0030 ff ff be c5 00 00 47 45 54 20 2f 20 48 54 54 50 .....GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 6e 65 76 65 /1.1..Ho st: neve
0050 72 73 73 6c 2e 63 6f 6d 0d 0a 55 73 65 72 2d 41 rssl.com ..User-A
0060 67 65 6e 74 3a 20 63 75 72 6c 2f 38 2e 32 31 2e gent: cu rl/8.21.
0070 30 0d 0a 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 0..Accep t: */*...
0080 0d 0a ..
```



File Edit View Go Capture

tcp.stream eq 120

No.	Time	Source
8413	62.519184400	10.8.0.1
8530	63.518978400	10.8.0.1
8771	65.521453400	10.8.0.1
9290	69.538408700	10.8.0.1
9308	69.835863700	34.2.0.1
9309	69.836622300	10.8.0.1
9310	69.837555000	10.8.0.1
9332	70.104959300	34.2.0.1
9333	70.104990700	34.2.0.1
9334	70.104996800	34.2.0.1
9335	70.105004500	34.2.0.1
9336	70.105413400	10.8.0.1
9337	70.105876500	10.8.0.1
9341	70.117230400	10.8.0.1

- > Frame 9310: Packet, 130 bytes
- > Ethernet II, Src: Intel Ethernet Adapter, Dst: 08:00:27:00:00:00
- > Internet Protocol Version 4, Src: 10.8.0.1, Dst: 34.2.0.1
- > Transmission Control Protocol, Src Port: 54434, Dst Port: 80, Seq: 304114400, Len: 104
- ✓ Hypertext Transfer Protocol
 - ✓ GET / HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /
 - Request Version: HTTP/1.1
 - Host: neverssl.com\r\n
 - User-Agent: curl/8.21.1\r\n
 - Accept: */*\r\n
 - \r\n
 - [Full request URI: http://neverssl.com/]

en_lab02.pcapng

Wireshark · Follow TCP Stream (tcp.stream eq 120) · cn_lab02.pcapng

GET / HTTP/1.1
Host: neverssl.com
User-Agent: curl/8.21.0
Accept: */*

HTTP/1.1 200 OK
Date: Sun, 02 Aug 2026 14:26:11 GMT
Server: Apache/2.4.66 ()
Upgrade: h2,h2c
Connection: Upgrade
Last-Modified: Wed, 25 Jun 2022 00:23:33 GMT
ETag: "f79-5e28b29d38e93"
Accept-Ranges: bytes
Content-Length: 3961
Vary: Accept-Encoding
Content-Type: text/html; charset=UTF-8

```
<html>  
  <head>  
    <title>NeverSSL - Connecting ... </title>  
    <style>  
      body {  
        font-family: Montserrat, helvetica, arial, sans-serif;  
        font-size: 16px;  
        color: #444444;  
        margin: 0;  
      }  
      h2 {  
        font-weight: 700;  
        font-size: 1.6em;  
        margin-top: 30px;
```

WS=256 SACK_PERM
WS=256 SACK_PERM
WS=256 SACK_PERM
wins=29200 len=1400
...5--\.....E-
t?00.....U...
.../P...AD...P...
.....GE T / HTTP
/1.1--Ho st: neve
rssl.com --User-A
gent: cu r1/8.21.
0--Accep t: */...
...
Mapped 0 (0.0%) Profiler Default
ENG IN 2003 02-08-2026

cn_lab02.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Wireshark · Conversations · cn_lab02.pcapng

Conversation Settings

- ☐ Name resolution
- ☐ Absolute start time
- ☐ Display raw data
- ☒ Limit to display filter

Copy

Follow Stream...

Graph...

I/O Graphs

Protocol

- ☐ Bluetooth
- ☐ BPv7
- ☐ DCCP
- ☐ DNP 3.0
- ☒ Ethernet
- ☐ FC

Filter list for specific type

Ethernet · 1 IPv4 · 1 IPv6 TCP · 1 UDP

Port A	Address B	Port B	Packets	Bytes	Stream ID	Total Packets	Percent Filtered	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration
61487	34.223.124.45	80	1	130 bytes	120	16	6.25%	1	130 bytes	0	0 bytes	62.519184400	7.865975

Accept: */*\r\n

\r\n

[Full request URI: http://neverssl.com/]

cn_lab02.pcapng

Packets: 16301 · Displayed: 1 (0.0%) Dropped: 0 (0.0%)

Profile Default

30°C
Mostly cloudyENG
IN2007
02-08-2026

Ques 3 Answer the following when you access the site, <http://neverssl.com>

A. How many HTTP GET requests do you observe?
List down the GET requests.

Ans Total HTTP GET Requests = 1
GET Request :- GET /HTTP/1.1

B. What version of HTTP the server is running?

Ans HTTP version = HTTP/1.1

C. For each of the HTTP GET requests, find out:-

i) The total number of TCP segments being received.

Ans, 3 TCP segments were received from the server.

ii) Total amount of data received in the corresponding HTTP response:-

Ans, Content length = 1514 byte

cn_lab02.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http

No.	Time	Source	Destination	Protocol	Length	Info
5620	39.238277000	10.85.20.254	104.192.109.75	HTTP	588	POST /commonquery HTTP/1.1 (application/x-www-form-urlencoded)
5621	39.240175200	10.85.20.254	104.192.109.75	HTTP	580	POST /commonquery HTTP/1.1 (application/x-www-form-urlencoded)
5664	39.511875400	104.192.109.75	10.85.20.254	TCP	1514	[TCP Previous segment not captured] 80 → 49175 [PSH, ACK] Seq=1461 Ack=930 Win=64228 Len=14
5665	39.511878900	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5667	39.511887400	104.192.109.75	10.85.20.254	HTTP	1514	[TCP Previous segment not captured] Continuation
5668	39.511889200	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5669	39.511891300	104.192.109.75	10.85.20.254	HTTP	1514	[TCP Out-Of-Order] Continuation
5670	39.511893200	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5671	39.511895600	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5672	39.511900900	104.192.109.75	10.85.20.254	HTTP	2974	Continuation
5688	39.533445400	104.192.109.75	10.85.20.254	HTTP	4434	[TCP Previous segment not captured] Continuation
5689	39.533447400	104.192.109.75	10.85.20.254	HTTP	1514	[TCP Out-Of-Order] Continuation
5690	39.533450700	104.192.109.75	10.85.20.254	HTTP	1514	Continuation
5709	39.787608100	104.192.109.75	10.85.20.254	HTTP	1514	Continuation

> Frame 5620: Packet, 588 bytes on wire (4704 bits), 588 bytes captured (4704 bits) on
> Ethernet II, Src: Intel_f2:f8:b9 (5c:80:b6:f2:f8:b9), Dst: ASIXElectron_35:2d:a0 (f8:00:00:00:00:00)
> Internet Protocol Version 4, Src: 10.85.20.254, Dst: 104.192.109.75

> Transmission Control Protocol, Src Port: 49175, Dst Port: 80, Seq: 1, Ack: 1, Len: 1
> Hypertext Transfer Protocol

> POST /commonquery HTTP/1.1\r\n

Host: spec.cloud.360safe.com\r\n

Accept: */*\r\n

Connection: Keep-Alive\r\n

Cache-Control: no-cache\r\n

> Content-Length: 342\r\n

Content-Type: application/x-www-form-urlencoded\r\n

\r\n

[Full request URI: http://spec.cloud.360safe.com/commonquery]

```
0070 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 43 6f 6e 41 63 63 65 70 74 3a 20 2a 2f 2a 0d 0a 43 6f 6e
0080 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 6c 6e 65 63 74 69 6f 6e 3a 20 4b 65 65 70 2d 41 6c
0090 69 76 65 0d 0a 43 61 63 68 65 2d 43 6f 6e 74 72 6f 6c 3a 20 6e 6f 2d 63 61 63 68 65 0d 0a 43 6f
00a0 6f 6c 3a 20 6e 6f 2d 63 61 63 68 65 0d 0a 43 6f 6e 67 74 68 3a 20 33 34 32 0d 0a 43 6f 6e 74 65
00b0 6e 74 65 6e 74 2d 4c 65 6e 67 74 68 3a 20 33 34 32 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a
00c0 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f 78 2d 77 77 77 2d 66 6f 72 6d 2d 75 72 6c 65 6e 63 6f 64
00d0 65 64 0d 0a 0d 0a 0e 06 24 21 04 08 ef 64 d5 18 00 00 09 00 f7 91 ec bd 45 aa cd ed 5f 54 51 3c
00e0 55 63 0d 19 02 b3 de 14 fe 3b e7 3c af 54 45 77 5d f8 b9 88 84 a0 17 d9 92 c5 c4 76 f7 dc c5 3d
00f0 03 47 07 a6 0e e6 c4 a2 8e 50 c8 a8 e4 fd 1b 6d 7a 8e 6e 56 b0 b6 3b 0e a6 84 90 f7 34 e9 b7 f6
0100 0100 00 09 00 f7 91 ec bd 45 aa cd ed 5f 54 51 3c 03 47 07 a6 0e e6 c4 a2 8e 50 c8 a8 e4 fd 1b 6d
0110 55 63 0d 19 02 b3 de 14 fe 3b e7 3c af 54 45 77 5d f8 b9 88 84 a0 17 d9 92 c5 c4 76 f7 dc c5 3d
0120 03 47 07 a6 0e e6 c4 a2 8e 50 c8 a8 e4 fd 1b 6d 7a 8e 6e 56 b0 b6 3b 0e a6 84 90 f7 34 e9 b7 f6
0130 0140 0150 e4 23 0e ef 48 63 8d 26 2d ac 9b 29 ad bc 8f da 7a 8e 6e 56 b0 b6 3b 0e a6 84 90 f7 34 e9 b7 f6
0140 0150 e4 23 0e ef 48 63 8d 26 2d ac 9b 29 ad bc 8f da 7a 8e 6e 56 b0 b6 3b 0e a6 84 90 f7 34 e9 b7 f6
0160 ca bc ec 55 d5 8b a3 68 55 f3 ff e2 ed 28 e6 29 ca bc ec 55 d5 8b a3 68 55 f3 ff e2 ed 28 e6 29
0170 84 33 d1 68 34 c1 2a 89 bc 0c bb 86 9f 2d 73 2c 84 33 d1 68 34 c1 2a 89 bc 0c bb 86 9f 2d 73 2c
0180 16 f2 ca df 65 9b ee 04 05 06 18 0c 17 42 b4 bb 16 f2 ca df 65 9b ee 04 05 06 18 0c 17 42 b4 bb
```

```
Accept: /*.*. Con
nection: Keep-Al
ive--Cac he-Contr
ol: no-c ache--Co
ntent-Le ngth: 34
2--Conte nt-Type:
applica tion/x-w
ww-form- urlencod
ed----- $!----
----- E... TQc
Uc----->-TEW
]-.....-v-
-G.....-P.....
z-nV-...-4...
-@-Hc-E-...-
-U...h U-...-
-3-h4-...-
-----e-...-B-
```

HTTP Accept (http.accept), 13 bytes

Packets: 16301 - Displayed: 62 (0.4%) - Dropped: 0 (0.0%)

Profile: Default

cn_lab02.pcapng

File Edit View Go Capture

tcp.stream eq 120

No.	Time	Source
8413	62.519184400	10.8
8530	63.518978400	10.8
8771	65.521453400	10.8
9290	69.538408700	10.8
9308	69.835863700	34.2
9309	69.836622300	10.8
9310	69.837555000	10.8
9332	70.104959300	34.2
9333	70.104990700	34.2
9334	70.104996800	34.2
9335	70.105004500	34.2
9336	70.105413400	10.8
9337	70.105876500	10.8
9341	70.117230400	10.8

> Frame 9310: Packet, 130 B

> Ethernet II, Src: Intel_F

> Internet Protocol Version

> Transmission Control Prot

> Hypertext Transfer Protoc

GET / HTTP/1.1\r\n

Request Method: GET

Request URI: /

Request Version: HT

Host: neverssl.com\r\n

User-Agent: curl/8.21.

Accept: */*\r\n

\r\n

[Full request URI: htt

Wireshark · Follow TCP Stream (tcp.stream eq 120) · cn_lab02.pcapng

GET / HTTP/1.1

Host: neverssl.com

User-Agent: curl/8.21.0

Accept: */*

HTTP/1.1 200 OK

Date: Sun, 02 Aug 2026 14:26:11 GMT

Server: Apache/2.4.66 ()

Upgrade: h2,h2c

Connection: Upgrade

Last-Modified: Wed, 29 Jun 2022 00:23:33 GMT

ETag: "f79-5e28b29d38e93"

Accept-Ranges: bytes

Content-Length: 3961

Vary: Accept-Encoding

Content-Type: text/html; charset=UTF-8

```
<html>
  <head>
    <title>NeverSSL - Connecting ... </title>
    <style>
      body {
        font-family: Montserrat, helvetica, arial, sans-serif;
        font-size: 16px;
        color: #444444;
        margin: 0;
      }
      h2 {
        font-weight: 700;
        font-size: 1.6em;
        margin-top: 30px;
      }
    </style>
  </head>
  <body>
  </body>
</html>
```

Packet 9334, 1 client pkt, 3 server pkts, 1 turn. Click to select.

Entire conversation (2877 bytes)

Show as ASCII

No delta times

Stream 120

Find:

☐ Case sensitive

Find Next

Filter Out This Stream

Print

Save as...

Back

Close

Help

Packet 9334, 1 client pkt, 3 server pkts, 1 turn. Click to select.

Entire conversation (2877 bytes)

Show as ASCII

No delta times

Stream 120

Find:

☐ Case sensitive

Find Next

Filter Out This Stream

Print

Save as...

Back

Close

Help

D. Identify TCP three way handshake used to establish the connection with the `http://neverssl.com`. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

Packet	Source - Destination	Sequence No.	ACK No.
SYN	10.85.20.254 - 34.223.124.45.80	seq = 0	0
SYN-ACK	34.223.124.45.80 - 10.85.20.254	seq = 0	ACK = 1
ACK	10.85.20.254 - 34.223.124.45.80	seq = 0	ACK = 1

cn_lab02.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn==1&tcp.port==80

No.	Time	Source	Destination	Protocol	Length	Info
8366	61.840797600	10.85.20.254	104.192.108.113	TCP	66	61485 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8381	62.127297300	104.192.108.113	10.85.20.254	TCP	62	80 → 61485 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=128
8413	62.519184400	10.85.20.254	34.223.124.45	TCP	66	61487 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8530	63.518978400	10.85.20.254	34.223.124.45	TCP	66	[TCP Retransmission] 61487 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8771	65.521453400	10.85.20.254	34.223.124.45	TCP	66	[TCP Retransmission] 61487 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9290	69.538406700	10.85.20.254	34.223.124.45	TCP	66	[TCP Retransmission] 61487 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
9308	69.835863700	34.223.124.45	10.85.20.254	TCP	60	80 → 61487 [SYN, ACK] Seq=0 Ack=1 Win=26883 Len=0 MSS=1460
11274	96.169859600	10.85.20.254	54.255.136.181	TCP	66	63837 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11288	96.256649800	54.255.136.181	10.85.20.254	TCP	60	80 → 63837 [SYN, ACK] Seq=0 Ack=1 Win=17922 Len=0 MSS=1460
11402	97.087008400	10.85.20.254	104.192.108.113	TCP	66	63838 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11435	97.355600400	104.192.108.113	10.85.20.254	TCP	62	80 → 63838 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=128
11749	100.870305800	10.85.20.254	104.192.108.113	TCP	66	63843 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
11777	101.154163700	104.192.108.113	10.85.20.254	TCP	62	80 → 63843 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 WS=128
11791	101.212116900	10.85.20.254	104.192.108.113	TCP	66	63845 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

> Frame 9308: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

> Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: Intel_f2:f8:b9 (54:00:05:1f:f2:f8:b9)

> Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.20.254

> Transmission Control Protocol, Src Port: 80, Dst Port: 61487, Seq: 0, Ack: 1, Len: 0

Source Port: 80

Destination Port: 61487

[Stream index: 120]

[Stream Packet Number: 5]

> [Conversation completeness: Complete, WITH_DATA (31)]

[TCP Segment Len: 0]

Sequence Number: 0 (relative sequence number)

Sequence Number (raw): 2245387390

[Next Sequence Number: 1 (relative sequence number)]

Acknowledgment Number: 1 (relative ack number)

0000 5c 80 b6 f2 f8 b9 f8 e4 3b 35 2d a0 08 00 45 00 \.....;5---E-

0010 00 2c 00 00 40 00 f2 06 ca 6c 22 df 7c 2d 0a 55 ., @... .1..|-U

0020 14 fe 00 50 f0 2f 85 d5 e4 7e a3 9e 41 4f 60 12 ...P./...~.AO

0030 69 03 30 f2 00 00 02 04 05 b4 00 00 1:0.....

cn_lab02.pcapng

Packets: 16301 · Displayed: 51 (0.3%) · Dropped: 0 (0.0%)

Profile: Default

30°C
Light rain

ENG
IN

20:00
02.08.2024